ZS-F9

Tetrahedral Self-Duality and the Hexagonal Mediation Structure

Polyhedral Origin of the (Z, X, Y) = (2, 3, 6) Sector Decomposition, Truncation Residue Exchange Operators, and the Algebraic Identity $A = \rho_X \cdot \rho_Y$

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§0. Abstract

The regular tetrahedron is the unique self-dual Platonic solid: its 4 vertices and 4 faces are exchanged under polyhedral duality. The Z-Spin framework (ZS-F2 v1.0) assigns the self-dual tetrahedron pair to the Z-sector mediator, but a precise derivation of how this self-duality controls the polyhedral mediation between the X-sector (truncated octahedron, tO) and Y-sector (truncated icosahedron, tI) has been left open. This paper closes that gap on three levels: representation-theoretic (§2–§3), combinatorial-cohomological (§4–§5), and operator-theoretic (§6).

First, the irreducible representations of the full tetrahedral symmetry group T_d have dimensions {1, 1, 2, 3, 3} whose squares sum to $|T_d| = 24$; the dimension-2 irrep E and the two dimension-3 irreps T_1 and T_2 reproduce exactly the Z-Spin sector dimensions (Z, X, Y) = (2, 3, 6) [DERIVED]. Second, the regular tetrahedron decomposes its 6 edges as $A_1 \oplus E \oplus T_2$ (dim 1+2+3 = 6), while its 4 vertices and 4 faces decompose identically as $A_1 \oplus T_2$; the difference dim(Edges) − dim(Vertices) = 2 = dim(E) = dim(Z) is the polyhedral source of Z-sector emergence [PROVEN]. Third, the hexagonal faces of tO (8) and tI (20), considered as representations of the common rotation subgroup $T = A_4$, decompose as $2\cdot 1 \oplus 2\cdot 3$ and $3\cdot 1 \oplus 1' \oplus 1'' \oplus 5\cdot 3$ respectively, yielding a T-equivariant Hom space of dimension 16 and a cokernel identity $F_{hex}(tI) - F_{hex}(tO) = F_{cut}(tI) = 12$ [DERIVED].

Fourth, and new in v1.0(Revised), the truncation-residue exchange theorem (§6): defining the normalized truncation residue $\rho_P = |V(tP) - F(tP)| / (V(tP) + F(tP))$ for a truncated polyhedron tP, the X-sector tO yields $\rho_X = 10/38 = 5/19 = \delta_X$, the Y-sector tI yields $\rho_Y = 28/92 = 7/23 = \delta_Y$, and their product is the geometric impedance

$$A = \rho_X \cdot \rho_Y = (5/19)(7/23) = 35/437,$$

identical to the corpus PROVEN identity $\delta_X \cdot \delta_Y = A$ (ZS-M6 §3.1 PROVEN). Self-duality of the tetrahedron forces $\rho_Z = 0$, ensuring A is the unique leading invariant. Combined with the corpus-PROVEN block constraint $L_{XY} \equiv 0$ (ZS-F1, ZS-S1) and the register-total normalization $\kappa^2 = A/Q = 35/4807$ (ZS-M6 §2.2 PROVEN), the Z-sector propagator reduces all X ↔ Y transfer to the residue-mode amplitudes $T^{(r)}_{X\to Y}(\mu) = (A/(Q\mu^2)) |r_Y\rangle\langle r_X|$, with second-order Schur complement corrections $\Delta L_X^{(Y)} = -(A/(Q\mu^2))^2 \chi_Y(\mu) |r_X\rangle\langle r_X|$ and $\Delta L_Y^{(X)} = -(A/(Q\mu^2))^2 \chi_X(\mu) |r_Y\rangle\langle r_Y|$

[DERIVED]. These four operator identities provide the explicit form of the heuristic claim that the X- and Y-sector "cut residues" exchange information through the self-dual Z-mediator.

Two PROVEN combinatorial identities support the framework: F(icosahedron) = F(octahedron) + F(dodecahedron), i.e. 20 = 8 + 12, and the Truncation-Dual Theorem applied to the tetrahedron itself yielding F(t-Tet) = 8 = 4 + 4, the unique Archimedean instance in which preserved faces and cut faces have the same count. Five hypotheses raised in earlier free-exploration sessions are explicitly RETRACTED in §7 in favor of the corrected statements; the v1.0(Revised) additions of §6 do not modify any v1.0 result.

Verification: 44 tests across 9 categories (Locked Inputs, T_d Irrep Structure, Tetrahedron V/F/E Decomposition, K_4 Spectrum, Truncation-Dual on Tetrahedron, Hexagonal Mediation under T, Hom Space Dimension, Combinatorial Identity, Residue Exchange Operator Identities), all PASS at machine precision via the companion script zs_f9_verify_v1_1.py. Zero new free parameters; all inputs LOCKED, PROVEN, or DERIVED in prior corpus papers.

Keywords: tetrahedral self-duality, T_d irreducible representations, sector decomposition, hexagonal mediation, T-equivariant cokernel, Z-sector emergence, truncation residue exchange, Schur complement, block Laplacian, polyhedral combinatorics, K_4 graph spectrum, zero free parameters.

Epistemic Status Legend

All claims in this paper are tagged with one of the following statuses, consistent with the Z-Spin v1.0 corpus convention:

Status	Definition
PROVEN	Mathematical theorem with complete proof under stated definitions, or numerical verification at machine precision ($\leq 10^{-10}$ residual).
DERIVED	Quantitative consequence of PROVEN items combined with Z-Spin axioms, with zero free parameters beyond A = 35/437.
RETRACTED	Claim previously advanced in free-exploration session that fails verification under rigorous calculation. Corrected statement provided.
NON-CLAIM	An explicit declaration that a specific quantity or interpretation is NOT derived from the framework.

§1. Introduction

1.1 The Self-Dual Tetrahedron in the Z-Spin Framework

The Z-Spin framework (ZS-F2 v1.0, PROVEN; ZS-F5 v1.0, PROVEN) assigns three distinct Archimedean polyhedra to the three sectors of its (Z, X, Y) = (2, 3, 6) decomposition: the truncated octahedron tO with octahedral symmetry O_h to the X-sector, the truncated icosahedron tI with

icosahedral symmetry I_h to the Y-sector, and the self-dual tetrahedron pair to the Z-sector mediator. The geometric impedance $A = 35/437 = \delta_X \cdot \delta_Y$ arises as a PROVEN algebraic identity from the duality-deviation invariants of the two truncated mediators (ZS-M6 §3.1, PROVEN).

The role of the tetrahedron in this assignment is structurally distinct from that of the other two mediators. Among the five Platonic solids, the regular tetrahedron is the unique self-dual one: its 4 vertices and 4 faces are exchanged by polyhedral duality with no change in count. Yet ZS-F2 v1.0 treats the tetrahedron as a Z-sector assignment without a precise representation-theoretic derivation of why self-duality forces $\dim(Z) = 2$ and how the tetrahedron's symmetry group T_d controls the X-Y mediation channels through its common subgroup relation with O_h and I_h, nor without an explicit operator-level statement of how truncation residues are exchanged between sectors.

1.2 Four Open Questions

Four specific structural questions about the tetrahedron's role have remained without explicit derivation in the v1.0 corpus:

(Q1) Why does the dimension $\dim(Z) = 2$ emerge specifically from the tetrahedron's geometry, rather than from an arbitrary choice consistent with the $j = 1/2$ SU(2) recoupling result of ZS-M3 Theorem 5.1 (PROVEN)?

(Q2) The Truncation-Dual Theorem $F(tP) = F(P) + F(P^*)$ (ZS-F2 §11.2 Theorem 11.2, PROVEN) gives $F(t\text{-}Tet) = 4 + 4 = 8$, the unique Archimedean instance in which preserved faces equal cut faces. What does this self-referential structure encode physically?

(Q3) The PROVEN combinatorial fact $F_{hex}(tI) = 20$ and $F_{hex}(tO) = 8$, combined with the cut-face count $F_{cut}(tI) = F(\text{dodecahedron}) = 12$, gives $20 - 8 = 12$. Is this difference structural, and how does it connect to the X-Y polyhedral mediation?

(Q4) The PROVEN identity $A = \delta_X \cdot \delta_Y$ (ZS-M6 §3.1) algebraically combines the X- and Y-sector duality-deviation invariants into the geometric impedance. What is the corresponding operator statement at the level of the block Laplacian, and how does the self-dual tetrahedron mediate the residue exchange that this identity encodes?

1.3 Four Answers and One Correction

This paper provides four structural answers, two combinatorial identities, and one explicit retraction:

(A1) The dimension $\dim(Z) = 2$ emerges as $\dim(E)$ where E is the unique 2-dimensional irreducible representation of T_d. Concretely, the 4 vertices and 4 faces of the regular tetrahedron decompose as $A_1 \oplus T_2$ (dim 1+3 = 4) under T_d, while the 6 edges decompose as $A_1 \oplus E \oplus T_2$ (dim 1+2+3 = 6). The difference $\dim(\text{Edges}) - \dim(\text{Vertices}) = 2$ is exactly $\dim(E)$ — the Z-sector representation. Self-duality ($V = F = 4$) preserves the $(A_1 \oplus T_2)$ summand, while the additional (E) summand is carried only by the edges. Z-sector emerges as the edge-exclusive irreducible representation of T_d.

(A2) The PROVEN identity $F(t\text{-}Tet) = F(Tet) + F(Tet^*) = 4 + 4$ is the unique case in which an Archimedean truncation produces equal numbers of preserved and cut faces. Consequently, the truncated tetrahedron's 8 faces split into 4 hexagons (from 4 preserved triangular faces) and 4 triangles (from 4 cut vertices = 4 faces of Tet*). This balanced 4 + 4 structure has no analog in the other four Archimedean truncations of Platonic solids.

(A3) The hexagonal mediation between tO and tI is controlled by the rotation subgroup $T = A_4$ (the largest common rotation subgroup of O and I in the standard alignment). Under T, the hexagonal face spaces decompose as $\Omega^2_{hex}(tO) = 2\cdot 1 \oplus 2\cdot 3$ (dim 8) and $\Omega^2_{hex}(tI) = 3\cdot 1 \oplus 1' \oplus 1'' \oplus 5\cdot 3$ (dim 20). The T-equivariant Hom space has dimension 16, a T-equivariant injection $tO \to tI$ exists, and its cokernel has dimension $20 - 8 = 12 = F_{cut}(tI) = F(\text{dodecahedron})$. The combinatorial identity $F(I) = F(O) + F(D)$, namely $20 = 8 + 12$, is the source of this cokernel-equals-cut relationship.

(A4) NEW in v1.0(Revised). Defining the normalized truncation residue $\rho_P = |V(tP) - F(tP)| / (V(tP) + F(tP))$ for a truncated polyhedron tP, one obtains $\rho_X = 5/19$, $\rho_Y = 7/23$, and crucially $\rho_Z = 0$ (forced by the tetrahedron's self-duality). The product identity $A = \rho_X \cdot \rho_Y$ is the corpus PROVEN identity $\delta_X \cdot \delta_Y = A$ (ZS-M6 §3.1). At the operator level, the block-Laplacian constraint $L_{XY} \equiv 0$ (ZS-F1, ZS-S1, PROVEN) combined with the register-total normalization $\kappa^2 = A/Q$ (ZS-M6 §2.2, PROVEN) yields the residue-exchange amplitude $T^{(r)}_{X\to Y}(\mu) = (A/(Q\mu^2)) |r_Y\rangle\langle r_X|$ and Schur complement sector corrections $\Delta L_X^{(Y)} = -(A/(Q\mu^2))^2 \chi_Y(\mu) |r_X\rangle\langle r_X|$, $\Delta L_Y^{(X)} = -(A/(Q\mu^2))^2 \chi_X(\mu) |r_Y\rangle\langle r_Y|$. The self-duality $\rho_Z = 0$ ensures that A is the leading mediation invariant; any non-self-dual Z-sector mediator would introduce an additional bias term and break the zero-free-parameter discipline. This is §6.

Correction. Five hypotheses raised in author preparatory notes and earlier free-exploration sessions are explicitly retracted in this paper. The full retraction record is given in §7. The corrected results above replace those hypotheses; verification at machine precision is provided in the companion script.

1.4 Locked Inputs and Notation

All inputs are LOCKED, PROVEN, or DERIVED in prior corpus papers. No new free parameters are introduced.

Table 1.1. Locked inputs from prior Z-Spin corpus.

Symbol	Value	Meaning	Source / Status
A	35/437	Geometric impedance	ZS-F2 v1.0, LOCKED
(Z, X, Y)	(2, 3, 6)	Sector decomposition	ZS-F5 v1.0, PROVEN
Q	11	Slot register dimension	ZS-F5 v1.0, PROVEN

Symbol	Value	Meaning	Source / Status
$\|T_d\|$	24	Full tetrahedral group order	Standard, PROVEN
$\|T\| = \|A_4\|$	12	Tetrahedral rotation subgroup	Standard, PROVEN
$\|O_h/T_d\|$	2	Cube/Octahedron coset count	Standard, PROVEN
$\|I_h/T_d\|$	5	Icosahedral coset count	ZS-U5 §5.3, PROVEN
$F(tP) = F(P) + F(P^*)$	—	Truncation-Dual Theorem	ZS-F2 §11.2, PROVEN
$L_{XY} \equiv 0$	—	Block Laplacian X-Y vanishes	ZS-F1, ZS-S1, PROVEN
$\delta_X = 5/19$, $\delta_Y = 7/23$	—	Sector duality-deviation invariants	ZS-M6 §3.1, PROVEN
$A = \delta_X \cdot \delta_Y = 35/437$	—	Algebraic factorization of A	ZS-M6 §3.1, PROVEN
$\kappa^2 = A/Q = 35/4807$	—	Register-Total Normalization	ZS-M6 §2.2, PROVEN

§2. T_d Irreducible Representations and Sector Correspondence

2.1 The Five Irreducible Representations of T_d

The full tetrahedral symmetry group T_d (order 24, isomorphic to S_4) admits five irreducible representations over the complex numbers. Their dimensions and the sum-of-squares identity are standard finite group theory.

Table 2.1. T_d irreducible representations and their characters.

Irrep	dim	8 C_3	3 C_2	6 S_4	6 σ_d	Z-Spin role
A_1	1	1	1	1	1	Trivial
A_2	1	1	1	−1	−1	Sign / chirality
E	2	−1	2	0	0	Z-sector irrep
T_1	3	0	−1	1	−1	Y-sector half
T_2	3	0	−1	−1	1	X-sector irrep

The dimension-squared sum verifies $|T_d|$:

$$1^2 + 1^2 + 2^2 + 3^2 + 3^2 = 1 + 1 + 4 + 9 + 9 = 24 = |T_d|. \quad (2.1)$$

This identity is PROVEN by the Wedderburn theorem applied to the group algebra $\mathbb{C}[T_d]$. The character orthogonality relations have been verified numerically (test set [B] of zs_f9_verify_v1_1.py).

2.2 The Sector Correspondence Theorem

Theorem 2.1 (Sector Correspondence). The dimensions of the irreducible representations of T_d coincide with the Z-Spin sector dimensions $(Z, X, Y) = (2, 3, 6)$ under the assignment

$$\dim(E) = 2 = \dim(Z), \quad \dim(T_2) = 3 = \dim(X), \quad \dim(T_1) + \dim(T_2) = 6 = \dim(Y). \quad (2.2)$$

Furthermore, the two trivial-dimensional irreducibles $\{A_1, A_2\}$ together carry the chirality structure $\dim(A_1) + \dim(A_2) = 2$, matching the Z-parity grading $\dim(Z) = 2$.

Proof. Direct comparison of standard T_d character table dimensions with PROVEN Z-Spin sector decomposition (ZS-F5 v1.0). The complexification $\mathbb{R}^3 \otimes_{\mathbb{R}} \mathbb{C} = \mathbb{C}^3$ used in ZS-M2 Corollary 4.1 (PROVEN) gives $Y = X \cdot Z = 3 \cdot 2 = 6$ multiplicatively, which corresponds at the T_d level to $T_1 \oplus T_2$ (dim $3 + 3 = 6$). ∎

[STATUS: DERIVED. Inputs: T_d character theory PROVEN; $(Z, X, Y) = (2, 3, 6)$ PROVEN ZS-F5 v1.0; $Y = X \cdot Z$ complexification DERIVED ZS-M2 Cor 4.1.]

2.3 Critical Distinction: Identification, not Derivation

Theorem 2.1 asserts that T_d's irreducible-representation dimensions reproduce the Z-Spin sector dimensions; it does not derive $(Z, X, Y) = (2, 3, 6)$ from group theory alone. The sector decomposition is established independently by ZS-F5 v1.0 §3 from the $Q = 11 = b_1 + \dim[SU(3)_{adj}]$ decomposition of the BCC T^3 Hodge complex (PROVEN), and by ZS-M3 Theorem 5.1 from the $j = 1/2$ uniqueness in 4-valent SU(2) recoupling (PROVEN). Theorem 2.1 records the consonance between these prior results and T_d representation theory; it is a structural identification at the level of dimensions, not a substitute for the prior derivations.

[NON-CLAIM NC-F9.1]: Theorem 2.1 does not claim that T_d's representation theory alone fixes the choice of $\dim(Z) = 2$ over alternative choices. The selection of the E irrep (rather than A_1 or T_2) for the Z-sector is forced by additional inputs: the $j = 1/2$ recoupling uniqueness (ZS-M3, PROVEN), the SU(2) double-cover structure of the seam involution (ZS-M3 Lemma 10.1, PROVEN), and the Truncation-Dual Theorem (PROVEN, this paper §3).

§3. Tetrahedron Vertex/Face/Edge Decomposition

3.1 The Three Combinatorial Spaces

The regular tetrahedron has 4 vertices, 4 faces, and 6 edges, related by Euler's formula $V - E + F = 4 - 6 + 4 = 2$. Under the action of T_d, each space carries a permutation representation, which decomposes into irreducible components by character theory.

3.2 Vertex and Face Representations: $A_1 \oplus T_2$

Lemma 3.1 (Vertex Representation). The 4 vertices of the regular tetrahedron, as a permutation representation of T_d, decompose as

$$V(Tet) \cong A_1 \oplus T_2, \quad \dim 1 + 3 = 4. \quad (3.1)$$

Proof. T_d acts transitively on the 4 vertices with vertex stabilizer C_{3v} (order 6). The induced permutation character is $\chi_{perm}(g) = \#\{\text{fixed vertices of } g\}$, evaluated on the five conjugacy classes (e, 8 C_3, 3 C_2, 6 S_4, 6 σ_d) as (4, 1, 0, 0, 2). Inner product with the T_d character table (Table 2.1) gives multiplicities (1, 0, 0, 0, 1) for (A_1, A_2, E, T_1, T_2). ▮

Lemma 3.2 (Face Representation = Vertex Representation). The 4 faces of the regular tetrahedron, as a permutation representation of T_d, decompose identically:

$$F(Tet) \cong A_1 \oplus T_2 \cong V(Tet). \quad (3.2)$$

Proof. By the Dual Face Lemma (ZS-F2 §11.2 Lemma 11.1, PROVEN), $V(P) = F(P^*)$ for any convex polyhedron P. The tetrahedron is self-dual: $Tet^* = Tet$. Therefore V(Tet) and F(Tet) carry isomorphic T_d representations. ▮

3.3 Edge Representation: $A_1 \oplus E \oplus T_2$

Theorem 3.3 (Edge Representation). The 6 edges of the regular tetrahedron, as a permutation representation of T_d, decompose as

$$E(Tet) \cong A_1 \oplus E \oplus T_2, \quad \dim 1 + 2 + 3 = 6. \quad (3.3)$$

Proof. T_d acts transitively on the 6 edges with edge stabilizer $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$ (order 4). The induced character on (e, 8 C_3, 3 C_2, 6 S_4, 6 σ_d) is (6, 0, 2, 0, 2). Inner product with the T_d character table gives multiplicities (1, 0, 1, 0, 1). ▮

3.4 The Z-Sector Emergence Theorem

Theorem 3.4 (Z-Sector Emergence). Under the T_d action on the regular tetrahedron, the difference of edge and vertex representations is exactly the 2-dimensional irreducible representation E:

$$E(Tet) - V(Tet) = (A_1 \oplus E \oplus T_2) - (A_1 \oplus T_2) = E. \quad (3.4)$$

Equivalently, dim(Edges) − dim(Vertices) = 6 − 4 = 2 = dim(E) = dim(Z). The Z-sector is the irreducible representation of T_d that appears in the edge space but not in the vertex (or face) space.

[STATUS: PROVEN. Direct character-theoretic computation; verified numerically in zs_f9_verify_v1_1.py tests C1–C4.]

Theorem 3.4 provides the polyhedral source of the Z-sector dimension. The self-duality V = F of the tetrahedron preserves the ($A_1 \oplus T_2$) summand under the V ↔ F exchange, while the additional (E) summand resides exclusively in the edge space. Self-duality and Z-sector emergence are two faces of the same algebraic structure. The same self-duality has an operator-level consequence at the level of truncation residues, made precise in §6 below: $\rho_Z = 0$.

3.5 K_4 Spectrum Realization

The regular tetrahedron's 1-skeleton is the complete graph K_4. The adjacency matrix $A_{K_4} = J_4 - I_4$ has spectrum

$$\sigma(A_{K_4}) = \{3, -1, -1, -1\}. \quad (3.5)$$

The eigenvalue 3 (multiplicity 1) corresponds to the constant eigenvector = trivial irrep A_1; the eigenvalue −1 (multiplicity 3) corresponds to T_2. The maximum eigenvalue $\lambda_{max} = 3 = \dim(X)$ reproduces the X-sector dimension at the spectral level.

[NON-CLAIM NC-F9.2]: The numerical equality $\lambda_{max}(K_4) = 3 = \dim(X)$ is a consequence of the T_d-equivariant decomposition $V(Tet) = A_1 \oplus T_2$ (Lemma 3.1, PROVEN). It is not an independent identity.

§4. Truncation-Dual Theorem on the Tetrahedron

4.1 The Self-Referential Truncation

The Truncation-Dual Theorem (ZS-F2 §11.2 Theorem 11.2, PROVEN) states that for any convex polyhedron P,

$$F(tP) = F(P) + F(P^*). \quad (4.1)$$

Applied to the icosahedron and octahedron, this gives $F(tI) = 20 + 12 = 32$ and $F(tO) = 8 + 6 = 14$. Applied to the self-dual tetrahedron, with $Tet^* = Tet$:

$$F(t\text{-}Tet) = F(Tet) + F(Tet^*) = 4 + 4 = 8. \quad (4.2)$$

The truncated tetrahedron has 8 faces, of which 4 are hexagons (from preserved triangular faces of Tet) and 4 are triangles (from cut vertices = faces of Tet*). This 4 + 4 partition is unique among the five Archimedean truncations of Platonic solids.

Table 4.1. Preserved/Cut face split for the five Archimedean truncations of Platonic solids.

t-P	F^pres	F^cut	F(tP)	F^pres − F^cut	Status
t-Tet	4	4	8	0	Self-referential
t-Cube	6	8	14	2	Asymmetric
t-Oct	8	6	14	−2	Asymmetric
t-Dod	12	20	32	8	Asymmetric
t-Ico	20	12	32	−8	Asymmetric

Observation 4.1 (Self-Referential Property). Among the five Archimedean truncations of Platonic solids, the truncated tetrahedron is the unique instance with $F^{pres}(tP) = F^{cut}(tP)$. This balance is forced by self-duality $V(Tet) = F(Tet)$ via the Truncation-Dual Theorem.

4.2 Polyhedral Identity F(I) = F(O) + F(D)

Lemma 4.2 (Cross-Pair Face Identity). Among the face counts of the five Platonic solids, there are exactly two non-trivial identities of the form $F(P_1) = F(P_2) + F(P_3)$ with all three solids distinct:

$$F(Dod) = F(Oct) + F(Tet), \quad 12 = 8 + 4 \quad (4.3)$$

$$F(Ico) = F(Dod) + F(Oct), \quad 20 = 12 + 8 \quad (4.4)$$

Proof. Direct enumeration over all $5 \times 4 \times 3 = 60$ ordered triples of distinct Platonic solids; verified in zs_f9_verify_v1_1.py. ▌

Identity (4.4) plays a structural role in §5: rewritten as $F_{hex}(tI) = F_{hex}(tO) + F_{cut}(tI)$, it gives the dimension count for the cokernel of the T-equivariant injection $\Omega^2_{hex}(tO) \to \Omega^2_{hex}(tI)$.

[NON-CLAIM NC-F9.3]: Identities (4.3) and (4.4) are PROVEN combinatorial facts about Platonic solid face counts. They are not derived from a deeper representation-theoretic principle in this paper.

§5. The Hexagonal Mediation Theorem

5.1 Setup: Common Rotation Subgroup $T = A_4$

The truncated octahedron tO carries O_h symmetry (order 48); the truncated icosahedron tI carries I_h symmetry (order 120). The largest common rotation subgroup acting simultaneously on both polyhedra in standard alignment is the tetrahedral rotation group $T \cong A_4$ (order 12). T sits inside $O = S_4$ with index 2 and inside $I = A_5$ with index 5 (ZS-U5 §5.3 PROVEN).

5.2 Hexagonal Face Decomposition under T

Table 5.1. Permutation characters under $T = A_4$ of the hexagonal face spaces.

Space	e	4 C_3	4 C_3^2	3 C_2	Status
$\Omega^2_{hex}(tO)$ (8 faces)	8	2	2	0	PROVEN
$\Omega^2_{hex}(tI)$ (20 faces)	20	2	2	0	PROVEN

Theorem 5.1 (T-Decomposition of Hexagonal Faces). Under the rotation subgroup $T = A_4$ (with complex irreducible representations 1, 1', 1'', 3), the hexagonal face spaces of tO and tI decompose as

$$\Omega^2_{hex}(tO) \cong 2 \cdot 1 \oplus 2 \cdot 3, \quad \dim 2(1) + 2(3) = 8. \quad (5.1)$$

$$\Omega^2_{hex}(tI) \cong 3 \cdot 1 \oplus 1' \oplus 1'' \oplus 5 \cdot 3, \quad \dim 3(1) + 1 + 1 + 5(3) = 20. \quad (5.2)$$

[STATUS: PROVEN. Numerical verification in zs_f9_verify_v1_1.py tests F4–F7 at machine precision.]

5.3 The Hexagonal Mediation Theorem

Theorem 5.2 (Hexagonal Mediation). The T-equivariant Hom space between the hexagonal face representations of tO and tI has complex dimension

$$\dim_{\mathbb{C}} \mathrm{Hom}_T(\Omega^2_hex(tO), \Omega^2_hex(tI)) = 16. \quad (5.3)$$

A T-equivariant injection $\Omega^2_hex(tO) \to \Omega^2_hex(tI)$ exists, and the cokernel of any such injection has complex dimension

$$\dim_{\mathbb{C}} \mathrm{coker} = 20 - 8 = 12 = F_cut(tI) = F(\text{dodecahedron}). \quad (5.4)$$

Proof. By Schur's lemma applied to T = A_4: dim Hom = (2)(3) + (0)(1) + (0)(1) + (2)(5) = 16. Injection conditions $m_target(\rho) \geq m_source(\rho)$ hold for all four irreps; cokernel multiplicities (1, 1, 1, 3) give complex dimension 1 + 1 + 1 + 9 = 12 = F(D) by Lemma 4.2. ▌

[STATUS: DERIVED. Verified in zs_f9_verify_v1_1.py tests G1–G3.]

5.4 Y-Exclusive Chirality Pair

Corollary 5.3 (Y-Exclusive Chirality Pair). The complex-conjugate pair {1', 1''} of T = A_4 appears in $\Omega^2_hex(tI)$ with total multiplicity 2 but is absent from $\Omega^2_hex(tO)$. The 2-dimensional Y-exclusive subspace is contained in the cokernel of any T-equivariant injection.

5.5 Cokernel-Cut Identity Restated

Combining Theorem 5.2 and Lemma 4.2:

$$F_hex(tI) - F_hex(tO) = F_cut(tI), \quad (5.6a)$$

$$F(I) = F(O) + F(D), \quad (5.6b)$$

$$\dim_{\mathbb{C}} \mathrm{coker}(\Omega^2_hex(tO) \to \Omega^2_hex(tI)) = F_cut(tI). \quad (5.6c)$$

[NON-CLAIM NC-F9.4]: Theorem 5.2 establishes a structural matching between the cokernel and the cut-face count F_cut(tI). It does NOT claim physical identification of the cokernel modes with dodecahedral faces — only dimensional agreement.

§6. Residue Exchange and the Operator Form of Mediation

[v1.0(Revised), April 2026]. This section provides the operator-theoretic counterpart to the representation-theoretic and combinatorial results of §3–§5. It states the algebraic identity $A = \rho_X \cdot \rho_Y$ in the form natural to the truncation-residue language, identifies the self-duality $\rho_Z = 0$ as the necessary condition for A to be the unique leading mediation invariant, and derives the Z-mediated residue-exchange amplitudes and Schur complement sector corrections from the corpus-PROVEN block-Laplacian structure. Throughout, all inputs are LOCKED, PROVEN, or DERIVED in prior corpus papers; no new free parameter is introduced.

6.1 Truncation Residue and Its Normalization

Definition 6.1 (Normalized Truncation Residue). For a Platonic solid P with truncation tP, define the truncation primal-dual residue

$$R(tP) := V(tP) - F(tP), \quad (6.1)$$

and the normalized truncation residue

$$\rho(tP) := |V(tP) - F(tP)| / (V(tP) + F(tP)). \quad (6.2)$$

This is the truncation-level analogue of the duality-deviation invariant $\delta(P) = |V(P) - F(P)| / (V(P) + F(P))$ introduced in ZS-F2 v1.0 §3 (PROVEN). The two coincide for the truncations relevant to the Z-Spin sector mediators, as shown in §6.2 below.

6.2 Sector Residues: $\rho_X = \delta_X$, $\rho_Y = \delta_Y$, $\rho_Z = 0$

Lemma 6.2 (Sector Residue Values). For the three sector mediators of the Z-Spin framework:

(i) X-sector — truncated octahedron tO with (V, E, F) = (24, 36, 14):

$$R(tO) = 24 - 14 = 10, \quad V + F = 38, \quad \rho_X = 10/38 = 5/19. \quad (6.3a)$$

(ii) Y-sector — truncated icosahedron tI with (V, E, F) = (60, 90, 32):

$$R(tI) = 60 - 32 = 28, \quad V + F = 92, \quad \rho_Y = 28/92 = 7/23. \quad (6.3b)$$

(iii) Z-sector — truncated tetrahedron t-Tet with (V, E, F) = (12, 18, 8). At the underlying Platonic level (the regular tetrahedron Tet itself, V = F = 4):

$$\rho_Z := |V(Tet) - F(Tet)| / (V(Tet) + F(Tet)) = 0/8 = 0. \quad (6.3c)$$

Furthermore, $\rho_X = \delta_X$ and $\rho_Y = \delta_Y$ (corpus PROVEN identifications, ZS-M6 §3.1).

Proof. Direct computation from the polyhedral data of tO and tI. The vanishing $\rho_Z = 0$ is a direct consequence of the tetrahedron's self-duality V(Tet) = F(Tet) = 4 (Lemma 3.2 of this paper, PROVEN). The identifications $\rho_X = \delta_X$ and $\rho_Y = \delta_Y$ follow from the algebraic identities $(V(tP) - F(tP)) / (V(tP) + F(tP)) = (V(P) - F(P)) / (V(P) + F(P)) = \delta(P)$ under the truncation map $(V, F) \mapsto (V', F') = (V, F + V)$ for a Platonic dual pair (P, P*); these reduce to $\delta_X = 5/19$ and $\delta_Y = 7/23$ (PROVEN, ZS-M6 §3.1). ▮

[STATUS: PROVEN. Verified in zs_f9_verify_v1_1.py tests I1, I2, I4.]

6.3 The Algebraic Identity $A = \rho_X \cdot \rho_Y$

Theorem 6.3 (Residue Product Identity). The geometric impedance A = 35/437 is the product of the two sector truncation residues:

$$A = \rho_X \cdot \rho_Y = (5/19) \cdot (7/23) = 35/437. \quad (6.4)$$

Equivalently, in raw polyhedral data:

$$A = [(V_X - F_X)(V_Y - F_Y)] / [(V_X + F_X)(V_Y + F_Y)] = (10 \cdot 28) / (38 \cdot 92) = 280/3496 = 35/437. \quad (6.5)$$

Proof. Direct algebraic identity from Lemma 6.2 combined with the corpus-PROVEN identity $A = \delta_X \cdot \delta_Y$ (ZS-M6 §3.1, PROVEN). ▮

[STATUS: PROVEN. Verified in zs_f9_verify_v1_1.py tests I3, I8 at exact rational arithmetic. This is a restatement of the corpus-PROVEN ZS-M6 §3.1 identity in the truncation-residue language.]

6.4 The Self-Duality Selection Principle

Corollary 6.4 (Self-Duality Selection). The vanishing $\rho_Z = 0$ ensures that the geometric impedance A is the unique leading mediation invariant of the X-Y polyhedral system. Any non-self-dual choice for the Z-sector mediator would yield $\rho_Z \neq 0$ and introduce an additional bias term, breaking the zero-free-parameter structure of the Z-Spin framework.

Proof. If $\rho_Z \neq 0$, the leading mediation invariant of the (X, Z, Y) triple would be a sum or product involving ρ_Z, e.g. $\rho_X \cdot \rho_Y + c \cdot \rho_Z$ for some structural coefficient c. The presence of ρ_Z would introduce an additional independent rational quantity into A, requiring at minimum one additional locked parameter or a tuned cancellation. Self-duality of the Z-mediator (V(Tet) = F(Tet)) collapses ρ_Z to 0 by Lemma 6.2(iii), eliminating this term. ▌

[STATUS: DERIVED. Inputs: Lemma 6.2 (PROVEN), zero-free-parameter discipline of ZS-F2 v1.0 (PROVEN). Provides the polyhedral selection rationale supporting ZS-F2's choice of the self-dual tetrahedron as Z-sector mediator.]

[NON-CLAIM NC-F9.5]: Corollary 6.4 articulates the structural rationale for assigning the self-dual tetrahedron to the Z-sector. It does not, by itself, prove that the only zero-residue Z-mediator candidate is the tetrahedron — one could in principle imagine other self-dual structures yielding $\rho_Z = 0$. The selection of the regular tetrahedron specifically is forced by additional inputs: the j = 1/2 SU(2) recoupling result (ZS-M3 Theorem 5.1, PROVEN) and the Z-sector emergence theorem (Theorem 3.4 of this paper, PROVEN).

6.5 Block Laplacian Structure and Z-Mediation

The Z-Spin block Laplacian (ZS-S1 v1.0 §4 PROVEN; ZS-Q1 v1.0 §2.2 PROVEN) takes the form

$$\mathcal{L}(\mu) = [[L_X + \mu^2 I_X, C_{XZ}, 0], [C_{ZX}, L_Z + \mu^2 I_Z, C_{ZY}], [0, C_{YZ}, L_Y + \mu^2 I_Y]] \quad (6.6)$$

with the corpus-PROVEN block constraint

$$L_{XY} \equiv 0 \quad \text{(PROVEN, ZS-F1 v1.0 §9, ZS-S1 v1.0 §4).} \quad (6.7)$$

This vanishing X-Y direct coupling is not an approximation; it follows from the $(1 + A\varepsilon^2)R$ coupling structure of the Z-Spin action, which generates X–Z and Z–Y intertwiners but no direct X–Y intertwiner. Consequently, all X ↔ Y transitions must factor through the Z-sector. Defining the Z propagator

$$G_Z(\mu) := (L_Z + \mu^2 I_Z)^{-1}, \quad (6.8)$$

the leading X→Y and Y→X effective transfer operators are

$$T_{X\to Y}(\mu) = C_{YZ} \cdot G_Z(\mu) \cdot C_{ZX}, \quad T_{Y\to X}(\mu) = C_{XZ} \cdot G_Z(\mu) \cdot C_{ZY}. \quad (6.9)$$

This factorization structure is the Z-Mediation Theorem of ZS-Q1 v1.0 §3.2 (PROVEN).

6.6 Residue-Mode Reduction and the $\kappa^2 = A/Q$ Coupling

Restricting to the rank-1 residue-mode approximation, denote the X- and Y-sector residue states $|r_X\rangle$ and $|r_Y\rangle$, and the physical Z-even mode $|z_0\rangle$ satisfying $L_Z|z_0\rangle = 0$ (ZS-S1 v1.0 §4 PROVEN: the Z-sector Z_2-even physical mode survives at $\beta_0(Z) = 1$, generating a rank-1 Schur correction). The intertwiners take the form

$$C_{ZX} \approx \kappa\, |z_0\rangle\langle r_X|, \quad C_{YZ} \approx \kappa\, |r_Y\rangle\langle z_0| \quad (6.10)$$

with the per-mode cross-coupling fixed by the Register-Total Normalization Theorem (ZS-M6 v1.0 §2.2, PROVEN) to

$$\kappa^2 = A / Q = 35/4807. \quad (6.11)$$

This is the same coupling that enters the Block Fiedler Mediation Theorem (ZS-T1 v1.0 §9.3 PROVEN), reciprocally governing the X-side fine-structure-constant face $1/\kappa^2 \approx 137$ and the Y-side solar Yukawa face $\kappa^2 \approx 0.0073$ of the Z-Spin spectral observatory.

6.7 Residue Exchange Amplitudes

Theorem 6.5 (Residue Exchange Amplitude). Under the rank-1 residue-mode approximation (6.10) with $\kappa^2 = A/Q$ (6.11), the leading X→Y and Y→X effective transfer operators take the form

$$T^{(r)}_{X\to Y}(\mu) = (A / (Q\, \mu^2))\, |r_Y\rangle\langle r_X|, \quad (6.12a)$$

$$T^{(r)}_{Y\to X}(\mu) = (A / (Q\, \mu^2))\, |r_X\rangle\langle r_Y|. \quad (6.12b)$$

Proof. Substituting (6.10) into (6.9) and using $L_Z|z_0\rangle = 0$:

$$\langle z_0|\, (L_Z + \mu^2 I_Z)^{-1}\, |z_0\rangle = 1/\mu^2. \quad (6.13)$$

Therefore

$$T^{(r)}_{X\to Y}(\mu) = \kappa^2 \cdot |r_Y\rangle\langle z_0| \cdot (1/\mu^2) \cdot |z_0\rangle\langle r_X| = (\kappa^2/\mu^2)\, |r_Y\rangle\langle r_X| = (A/(Q\mu^2))\, |r_Y\rangle\langle r_X|.$$

The Y→X amplitude follows by Hermitian conjugation. ▮

[STATUS: DERIVED. Inputs: $L_{XY} \equiv 0$ PROVEN (ZS-F1, ZS-S1); Z-Mediation Theorem PROVEN (ZS-Q1 §3.2); rank-1 Schur structure PROVEN (ZS-S1 §4); $\kappa^2 = A/Q$ PROVEN (ZS-M6 §2.2). Verified in zs_f9_verify_v1_1.py tests I5, I6 at machine precision via 3×3 toy block.]

Theorem 6.5 is the operator-theoretic counterpart to Theorem 6.3. The algebraic identity $A = \rho_X \cdot \rho_Y$ of (6.4) now appears as the coefficient of the rank-1 residue-exchange operator (6.12); the polyhedral data of the cut residues (10/38, 28/92) are encoded operator-theoretically in the X→Y and Y→X transfer amplitudes.

6.8 Sector Corrections from Schur Complement

Theorem 6.6 (Schur Sector Corrections). The leading second-order Schur complement corrections to the X- and Y-sector effective Laplacians, induced by integrating out the complementary sector through the Z-mediator, take the rank-1 form

$$\Delta L_X^{(Y)}(\mu) = -(A / (Q\, \mu^2))^2 \cdot \chi_Y(\mu) \cdot |r_X\rangle\langle r_X|, \quad (6.14a)$$

$$\Delta L_Y^{(X)}(\mu) = -(A / (Q\, \mu^2))^2 \cdot \chi_X(\mu) \cdot |r_Y\rangle\langle r_Y|, \quad (6.14b)$$

where the propagator susceptibilities are

$$\chi_Y(\mu) := \langle r_Y| (L_Y + \mu^2 I_Y)^{-1} |r_Y\rangle, \quad \chi_X(\mu) := \langle r_X| (L_X + \mu^2 I_X)^{-1} |r_X\rangle. \quad (6.15)$$

Proof. Standard Schur complement: integrating out Y from (6.6) yields

$$L_X^{eff} = L_X + \mu^2 I_X - C_{XZ} \cdot [L_Z + \mu^2 I_Z - C_{ZY} \cdot G_Y \cdot C_{YZ}]^{-1} \cdot C_{ZX}.$$

Expanding to leading order in $C_{ZY} \cdot G_Y \cdot C_{YZ}$ (a perturbation of the Z propagator) gives

$$\Delta L_X^{(Y)} = -C_{XZ} \cdot G_Z \cdot C_{ZY} \cdot G_Y \cdot C_{YZ} \cdot G_Z \cdot C_{ZX}.$$

Substituting (6.10) and (6.13) gives (6.14a). The symmetric formula (6.14b) follows by exchanging X and Y. ▌

[STATUS: DERIVED. Standard Schur complement applied to the Z-Spin block-Laplacian; inherits PROVEN structure of (6.6)–(6.11). Verified at the toy-block level in zs_f9_verify_v1_1.py test I6.]

6.9 Physical Interpretation

Theorems 6.3, 6.5, and 6.6 together provide the operator-theoretic statement of the X-Y mediation:

(i) The geometric impedance $A = \rho_X \cdot \rho_Y$ is the algebraic invariant carried by the cut-portion residues of the two sector mediators (Theorem 6.3, restating ZS-M6 §3.1 PROVEN).

(ii) The self-duality $\rho_Z = 0$ of the Z-mediator ensures that A is the unique leading invariant; any non-self-dual mediator would introduce a bias term and break zero-free-parameter discipline (Corollary 6.4).

(iii) At the operator level, the cut residues exchange information between sectors via the residue-exchange amplitudes $T^{(r)}_{X\to Y}$, $T^{(r)}_{Y\to X}$ (Theorem 6.5), with coefficient $A/(Q\mu^2)$ determined entirely by the locked constants A and Q.

(iv) The complementary sector induces second-order Schur complement corrections $\Delta L_X^{(Y)}$, $\Delta L_Y^{(X)}$ on the residue-mode subspaces, with coefficient $(A/(Q\mu^2))^2$ and shape determined by the sector susceptibilities $\chi_X(\mu)$, $\chi_Y(\mu)$ (Theorem 6.6).

Heuristic translation. The cut portions of the X-sector polyhedron (10 residual primal-dual asymmetry units of tO) and of the Y-sector polyhedron (28 residual primal-dual asymmetry units of tI) exchange information through the self-dual Z-mediator (residue 0). The product $(10 \cdot 28) / (38 \cdot 92) = 35/437$ is the algebraic content of this exchange; the operator $(A/(Q\mu^2)) |r_Y\rangle\langle r_X|$ is its rank-1 form on the residue mode subspace.

[NON-CLAIM NC-F9.6]: The rank-1 residue-mode approximation (6.10) is a leading-order approximation that captures the dominant Z-mediated transfer. Higher-rank corrections to C_{ZX}, C_{YZ} involving non-residue modes of the X- and Y-sectors are present at subleading order and are not derived in this paper. The full mode structure of these corrections is the subject of ZS-M6 v1.0 §2 (PROVEN at spectral level) and ZS-S1 v1.0 §4 (PROVEN at action level).

§7. Retractions: Five Hypotheses Withdrawn

Five hypotheses were advanced in author preparatory notes and earlier free-exploration sessions. Under rigorous numerical verification (zs_f9_verify_v1_1.py), these hypotheses fail at the stated level of strength and are explicitly retracted in this paper. The corrected statements are those given in §3–§6 above.

Table 7.1. Retracted hypotheses and corrected statements.

ID	Original Hypothesis	Verified Status / Correction
RH-F9.1	Ω^2_hex(tO) decomposes under T_d as A_1 ⊕ A_2 ⊕ T_1 ⊕ T_2 (dim 8).	Replaced by Theorem 5.1: under T = A_4, decomposition is 2 · 1 ⊕ 2 · 3.
RH-F9.2	A_2 chirality irrep is present in Ω^2_hex(tO) with multiplicity 1 and is the source of X→Y chirality loss.	RETRACTED. m_{A_2}(tO) computed = 0. The chirality structure is instead carried by the 1' + 1" pair exclusively in tI (Corollary 5.3).
RH-F9.3	dim Hom_{T_d}(Ω^2_hex(tO), Ω^2_hex(tI)) = 8 = F_hex(tO).	RETRACTED. Correct dim Hom_T = 16 (Theorem 5.2).
RH-F9.4	F_hex(tI) − dim Hom = 12 = F_cut(tI).	RETRACTED in this form. Correct identity is F_hex(tI) − F_hex(tO) = F_cut(tI), which equals the cokernel of injection (5.6c).
RH-F9.5	X cut faces (6 squares) and Y cut faces (12 pentagons) are exchanged in a 1:2 ratio matching ZS-Q7 Theorem 1's Γ(X→Y)/Γ(Y→X) = 2.	RETRACTED as a derivation. Reduced to OBSERVATION: the numerical ratio 12/6 = 2 matches Γ(X→Y)/Γ(Y→X) = 2 (PROVEN ZS-Q7), but the cut-face direct exchange is forbidden by ZS-M1 §8 (C_4 ⊂ O_h, C_5 ⊂ I_h are mutually exclusive). The 1:2 ratio is therefore a consistency check, not a causal mechanism.

These retractions are recorded in full as part of the paper's verification discipline. Self-correction of OBSERVATION-level free-exploration claims via rigorous numerical verification is a constitutive practice of the Z-Spin framework.

§8. Falsification Gates

Seven falsification gates are pre-registered for ZS-F9 v1.0(Revised): five from v1.0, plus two new gates from §6.

Table 8.1. Pre-registered falsification gates for ZS-F9 v1.0(Revised).

Gate	Falsification Condition	Consequence	Timeline
F-F9.1	Independent recomputation of T_d character table inner products yields different multiplicities in Lemmas 3.1, 3.2, 3.3 at integer level.	Theorems 3.1–3.4 falsified.	Immediate (verify script)

Gate	Falsification Condition	Consequence	Timeline
F-F9.2	T-equivariant Hom dimension between Ω^2_hex(tO) and Ω^2_hex(tI), computed at machine precision, differs from 16 by more than 10^{-10}.	Theorem 5.2 falsified.	Immediate (verify script)
F-F9.3	The T-decomposition multiplicities of Theorem 5.1 are sensitive to the choice of T ⊂ I_h embedding (different conjugates yield different multiplicities).	Theorem 5.1 demoted to T-embedding-dependent.	OPEN (analytic)
F-F9.4	The combinatorial identity F(I) = F(O) + F(D) admits a structural derivation from a deeper principle, making Lemma 4.2 a theorem rather than a fact.	Lemma 4.2 promoted; NC-F9.3 retracted.	OPEN (research)
F-F9.5	Theorem 3.4 (Z-sector emergence as edge-exclusive E irrep) extends to other Platonic solids in a non-trivial way that conflicts with dim(Z) = 2.	Theorem 3.4 weakened to tetrahedron-specific.	Immediate (verify script extension)
F-F9.6	Independent toy-block evaluation of the effective L_XY^eff coupling (after integrating out Z) yields a value differing from $-A/(Q\mu^2)$ by more than 10^{-10} at machine precision.	Theorem 6.5 falsified.	Immediate (verify script test I6)
F-F9.7	An alternative non-self-dual Z-mediator candidate is exhibited that produces ρ_Z = 0 by some other mechanism, demonstrating that self-duality is not necessary for the zero-residue condition.	Corollary 6.4 weakened from selection rule to consistency check.	OPEN (combinatorial enumeration)

Gates F-F9.1, F-F9.2, F-F9.5, F-F9.6 are immediate-verification gates with PASS status confirmed by zs_f9_verify_v1_1.py. Gates F-F9.3, F-F9.4, F-F9.7 are OPEN structural questions for future work.

§9. Conclusion

This paper has established four structural results connecting the self-dual tetrahedron to the Z-Spin sector decomposition (Z, X, Y) = (2, 3, 6). First, the dimensions of T_d's irreducible representations {1, 1, 2, 3, 3} reproduce the sector dimensions exactly, with the 2-dimensional irrep E corresponding to the Z-sector (Theorem 2.1). Second, the regular tetrahedron's edge representation A_1 ⊕ E ⊕ T_2 contains the E irrep, while its vertex and face representations (which are equal by self-duality) decompose as A_1 ⊕ T_2 — the difference is exactly the Z-sector's 2-dimensional irrep (Theorem 3.4). Third, the hexagonal face spaces of tO and tI, considered as representations of the common rotation subgroup T = A_4, give a T-equivariant Hom space of dimension 16, with cokernel dimension 20 − 8 = 12 = F_cut(tI), matching the polyhedral identity F(I) = F(O) + F(D) (Theorem 5.2). Fourth, and at the operator level, the truncation residues ρ_X = δ_X = 5/19 and ρ_Y = δ_Y = 7/23 of the two sector mediators combine into the geometric impedance A = ρ_X · ρ_Y = 35/437 (Theorem 6.3), with self-duality forcing ρ_Z = 0 (Corollary 6.4) and the block-Laplacian structure giving rank-1 residue exchange amplitudes T^(r)_X↔Y(μ) = (A/(Qμ²)) |r_·⟩⟨r_·| (Theorem 6.5) and Schur sector corrections ΔL_X^(Y), ΔL_Y^(X) (Theorem 6.6).

The four results are mutually reinforcing. Theorem 2.1 provides the dimensional skeleton; Theorem 3.4 provides the polyhedral source of dim(Z) = 2; Theorem 5.2 provides the X-Y hexagonal mediation cokernel structure; and §6 provides the operator-level realization of the X-Y residue exchange. Self-duality of the tetrahedron is the common thread: it forces V(Tet) = F(Tet) (Lemma 3.2), which in turn forces $\rho_Z = 0$ (Lemma 6.2(iii)), which selects $A = \rho_X \cdot \rho_Y$ as the unique leading invariant (Corollary 6.4), and edge-exclusively carries the 2-dimensional E irrep (Theorem 3.4) — the Z-sector representation that mediates the X-Y operator transfer.

Five hypotheses raised in earlier free-exploration sessions are explicitly retracted in §7, and seven falsification gates are pre-registered in §8. The structural content of this paper is consonant with the existing Z-Spin v1.0 corpus: no prior result is modified; no new free parameter is introduced. Two open structural questions remain (gates F-F9.3, F-F9.4, F-F9.7); their resolution would strengthen the framework but is not required for the present results.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude) for mathematical verification, character-theoretic computation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

The verification suite is publicly available.

Verification script: zs_f9_verify_v1_1.py.
Dependencies: Python 3.10+, numpy, mpmath.
Execution: python3 zs_f9_verify_v1_1.py
Expected output: 44/44 PASS, exit code 0.

Appendix A. Verification Suite Results

All 44 tests pass at machine precision.

Table A.1. Verification suite results for zs_f9_verify_v1_1.py.

Category	Content	Tests	Pass/Fail
[A] Locked Inputs	A = 35/437; (Z,X,Y) = (2,3,6); $\lvert O_h/T_d\rvert = 2$; $\lvert I_h/T_d\rvert = 5$; $\lvert T_d\rvert = 24$, $\lvert T\rvert = 12$.	5	5/0
[B] T_d Irrep Structure	Sum dim² = 24; E ↔ Z; T_2 ↔ X; T_1 + T_2 ↔ Y; A_1 + A_2 = 2 trivial.	5	5/0
[C] Tet V/F/E Decomposition	V = A_1 + T_2 (4); F = A_1 + T_2 (4); E = A_1 + E + T_2 (6); Edges − Vertices = 2.	4	4/0
[D] K_4 Spectrum	Eigenvalues {3, −1, −1, −1}; spectral gap 4; trace(K_4^2)/2 = 6.	4	4/0

Category	Content	Tests	Pass/Fail
[E] Truncation-Dual on Tet	F(Tet*) = F(Tet) = 4; F(t-Tet) = 8 = 4 + 4; preserved = cut.	3	3/0
[F] Hexagonal Mediation	$\|T\| = 12$; F(O) = 8, F(I) = 20; T classes (1, 4, 4, 3); $\chi_{tO} = (8,2,2,0)$; $\chi_{tI} = (20,2,2,0)$; decompositions $2\cdot1+2\cdot3$ and $3\cdot1+1'+1''+5\cdot3$; cokernel identity.	8	8/0
[G] Hom Space Dimension	dim $\mathrm{Hom}_T = 16$; injection exists; cokernel = 12 = $F_{cut}(tI)$; chirality pair Y-exclusive.	4	4/0
[H] Combinatorial Identity	F(I) = F(O) + F(D): 20 = 8 + 12; F_{hex} partition; F(I)/F(O) = 5/2 = $\|I_h\|/\|O_h\|$.	3	3/0
[I] Residue Exchange	$\rho_X = 5/19 = \delta_X$; $\rho_Y = 7/23 = \delta_Y$; $A = \rho_X\cdot\rho_Y = 35/437$; $\rho_Z = 0$; $\kappa^2 = A/Q = 35/4807$; toy-block $L_{XY}^{eff} = -A/(Q\mu^2)$; bare $L_{XY} = 0$; algebraic exactness.	8	8/0
TOTAL	All categories PASS at machine precision.	44	44/0

Appendix B. Cross-Reference Table

Table B.1. Dependencies and cross-references.

Result	Status	Source / Dependency
A = 35/437 (geometric impedance)	LOCKED	ZS-F2 v1.0 §5
(Z, X, Y) = (2, 3, 6); Q = 11	PROVEN	ZS-F5 v1.0 §3, §4
$A = \delta_X \cdot \delta_Y$ (algebraic identity)	PROVEN	ZS-M6 v1.0 §3.1
$\kappa^2 = A/Q$ (Register-Total Normalization)	PROVEN	ZS-M6 v1.0 §2.2
Truncation-Dual Theorem	PROVEN	ZS-F2 v1.0 §11.2 Theorem 11.2
$L_{XY} \equiv 0$ block Laplacian	PROVEN	ZS-F1 v1.0 §9, ZS-S1 v1.0 §4
Z-Mediation Theorem (operator factorization)	PROVEN	ZS-Q1 v1.0 §3.2 Theorem 3.1
j = 1/2 SU(2) recoupling uniqueness	PROVEN	ZS-M3 v1.0 Theorem 5.1
$\|I_h/T_d\| = 5$	PROVEN	ZS-U5 v1.0 §5.3
Block Fiedler Mediation Theorem	PROVEN	ZS-T1 v1.0 §9.3
F(t-Tet) = 8 = 4 + 4	PROVEN	This paper §4 + ZS-F2 §11.2
E(Tet) − V(Tet) = E (Z-sector)	PROVEN	This paper §3 Theorem 3.4
Hexagonal Mediation Theorem	DERIVED	This paper §5 Theorem 5.2
$A = \rho_X \cdot \rho_Y$ (residue product)	PROVEN	This paper §6 Theorem 6.3 (= ZS-M6 §3.1)
$\rho_Z = 0$ self-duality selection	DERIVED	This paper §6 Corollary 6.4

Result	Status	Source / Dependency
$T^{(r)}_{X\to Y} = (A/(Q\mu^2)) \lvert r_Y\rangle\langle r_X\rvert$	DERIVED	This paper §6 Theorem 6.5
$\Delta L_X^{(Y)}$, $\Delta L_Y^{(X)}$ Schur corrections	DERIVED	This paper §6 Theorem 6.6

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Version History

v1.0 (April 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes through April 2026.) Contents: Theorems 2.1, 3.1–3.4, 4.1, 5.1–5.3 with Corollary 5.3; Lemma 4.2 and Observation 4.1; five retractions RH-F9.1 through RH-F9.5; five falsification gates F-F9.1 through F-F9.5; verification suite zs_f9_verify_v1_0.py with 36/36 PASS at machine precision; four NON-CLAIMs NC-F9.1 through NC-F9.4.

v1.0(Revised) (April 2026): NEW §6 "Residue Exchange and the Operator Form of Mediation." Adds Definition 6.1 (Normalized Truncation Residue), Lemma 6.2 (Sector Residue Values), Theorem 6.3 (Residue Product Identity $A = \rho_X \cdot \rho_Y$, restating ZS-M6 §3.1 PROVEN in truncation-residue language), Corollary 6.4 (Self-Duality Selection), Theorem 6.5 (Residue Exchange Amplitude $T^{(r)}_{X\leftrightarrow Y} = (A/(Q\mu^2)) \lvert r_\cdot\rangle\langle r_\cdot\rvert$), and Theorem 6.6 (Schur Sector Corrections $\Delta L_X^{(Y)}$, $\Delta L_Y^{(X)}$). New non-claims NC-F9.5 (self-duality is necessary but not sufficient to fix Tet as Z-mediator) and NC-F9.6 (rank-1 residue-mode approximation is leading-order). New falsification gates F-F9.6 (toy-block effective coupling) and F-F9.7 (alternative $\rho_Z = 0$ mediators). Verification suite extended to 44/44 PASS via Category [I] (8 new tests). Section numbering shifted: previous §6 (Retractions) → §7; previous §7 (Falsification Gates) → §8; previous §8 (Conclusion) → §9. No prior content deleted; v1.0 results preserved verbatim. The v1.0(Revised) additions integrate Kenny's truncation-residue intuition (drafted

prior to this paper) with the corpus-PROVEN block-Laplacian structure, making explicit the operator-theoretic content of the algebraic identity $A = \delta_X \cdot \delta_Y$.